Iterative Self-Refinement for Multi-Camera Calibration via Learned Pose Conditioning

Abstract

Multi-camera extrinsic calibration fails in wide-baseline deployments where dynamic, visually repetitive content dominates the scene and static geometry is degenerate. Feed-forward geometry models such as Pi3X can estimate poses in these settings, but their accuracy saturates under standard post-processing. We show that feeding a model's own predictions back as pose conditioning creates an iterative refinement loop with a surprising asymmetry: rotation contracts toward the correct solution on SO(3) (contraction rate $\lambda \approx 0.8$), while translation diverges due to Sim(3) scale ambiguity. To exploit this, we propose *decoupled hybrid fusion*—sourcing rotation edges from conditioned inference and translation edges from unconditioned inference within a unified pose graph. Combined with Monte Carlo perturbation of the conditioning signal, this achieves 28% rotation improvement and 21% relative pose improvement over the raw model output on a challenging 8-camera feedlot dataset, with no external calibration data required. We derive the method's accuracy floor from the contraction rate and validate the prediction experimentally.

1 Introduction

Extrinsic calibration of wide-baseline multi-camera systems is a prerequisite for 3D reconstruction, tracking, and scene understanding. Classical approaches rely on either physical calibration targets [?] or structure-from-motion pipelines [?] that extract and match distinctive static features across views. Both strategies fail when cameras are spaced far apart with minimal field-of-view overlap, when the dominant visual content is dynamic and repetitive, and when the static scene geometry is degenerate. These three conditions co-occur in a growing class of deployments—livestock monitoring, wildlife observation, sports analytics, and large-area surveillance—where cameras observe moving subjects of near-identical appearance over geometrically unstructured backgrounds.

We study this problem in the concrete setting of eight wide-baseline cameras monitoring a cattle feedlot. The scene exhibits all three failure modes simultaneously: cameras are separated by 5–10 m with narrow overlap, each frame is dominated by moving cattle of near-identical black appearance, and the static pen is an approximately planar surface with few distinctive landmarks. Classical SfM pipelines produce no usable output in this setting.

Recent feed-forward geometry models [?, ?, ?] bypass explicit feature correspondence by learning to infer multi-view geometry directly from images. Pi3X [?], a permutation-equivariant model that processes all N views jointly, produces camera poses with $\sim 4.4^\circ$ mean rotation error from raw images alone—succeeding where classical methods fail entirely. However, standard geometric post-processing (robust averaging, pose graph optimization) yields no further rotation improvement, because the residual error is systematic bias from weak geometric signal rather than statistical noise amenable to averaging.

We observe that Pi3X accepts optional pose conditioning: feeding approximate camera poses alongside images improves its predictions. This raises the natural question of whether the model can refine *its own output* through iterative self-conditioning—analogous to fixed-point iteration on the pose manifold. We discover a fundamental asymmetry in this process: **self-conditioning contracts rotation toward the correct solution but causes translation to diverge.** The divergence arises because Pi3X's outputs are defined only up to Sim(3)—rotation is scale-invariant and thus stable under re-injection, while translation accumulates compounding scale bias when fed back as SE(3) conditioning.

This asymmetry motivates two questions that we address in this paper: *Under what conditions does self-conditioning converge on the rotation manifold?* And: *How can the rotation–translation asymmetry be exploited to improve calibration without degrading either component?*

Our contributions are as follows:

1. We provide the first empirical characterisation of self-conditioning dynamics in learned geometry models, showing that conditioned inference acts as a contraction mapping on SO(3) with measurable rate $\lambda \approx 0.8$, while the translation component does not contract due to Sim(3) scale ambiguity.

2. We propose *decoupled hybrid fusion*, a method that exploits this asymmetry by sourcing rotation edges from conditioned inference and translation edges from unconditioned inference within a unified pose graph.

3. We introduce a Monte Carlo perturbation strategy that interprets conditioned inference as approximate posterior sampling, and show that iterating hybrid fusion with stochastic perturbation achieves 28% rotation improvement and 21% relative pose improvement over the baseline without any external calibration data.

4. We derive the method's accuracy floor from the measured contraction rate and validate the prediction experimentally, establishing principled criteria for when self-refinement helps and when it saturates.

2 Related Work

Learned multi-view geometry. Feed-forward models for multi-view 3D reconstruction have advanced rapidly. DUSt3R [?] introduced pairwise pointmap regression; MASt3R [?] extended it with local feature matching. Pi3X [?] and VGGT [?] process all N views jointly via permutation-equivariant architectures, achieving stronger global consistency. These models output camera poses and dense geometry from images alone, but their accuracy on wide-baseline, feature-adversarial scenes remains limited.

Multi-camera calibration. Classical extrinsic calibration uses physical targets (checkerboards [?]), vanishing points [?], or structure-from-motion pipelines [?]. Target-free methods assume sufficient static texture for feature matching across views—an assumption violated in dynamic, textureless environments. Our setting falls outside the scope of these methods.

Pose graph optimization. Rotation averaging [?, ?] and translation averaging [?] are classical tools for aggregating noisy pairwise relative poses into a globally consistent estimate. We adopt these as our post-processing backbone (Stage 1) and as the fusion mechanism for combining heterogeneous edge sources (Stages 3–4).

Test-time refinement. Test-time training (TTT) [?] and test-time adaptation adapt model predictions at inference time using self-supervised objectives. In geometry estimation, self-supervised depth refinement [?] fine-tunes networks on test sequences using photometric consistency. Our approach differs in that we do *not* modify model weights—instead, we exploit the model's conditioning interface to iteratively refine predictions, treating the pretrained model as a fixed denoiser.

Iterative pose refinement. Classical approaches such as ICP [?] and PnP+RANSAC refinement iterate between correspondence estimation and pose optimization. Bundle adjustment [?] jointly optimises poses and 3D structure. Our method is conceptually similar—iterative refinement toward a fixed point—but operates through a learned model's conditioning mechanism rather than explicit geometric optimization.

Denoising and diffusion on manifolds. Score-based diffusion models [?] learn to denoise data via iterative refinement from noise toward the data manifold. Recent work has extended diffusion to SE(3) [?] for molecular and pose generation. Our Monte Carlo perturbation strategy (Section **??**) shares this structure: perturb the current estimate, apply a learned denoiser (Pi3X conditioning), and average. However, we operate in a discriminative setting (fixed pretrained model, no diffusion training) and exploit the observation that convergence is restricted to the SO(3) component.

3 Foundation Geometry Model Benchmark

To select the best backbone for our calibration pipeline, we benchmark seven recent foundation geometry models on our dataset. All models receive the same 8 synchronised camera images per timestep and must output camera poses without any prior information.

3.1 Experimental Protocol

We sample 91 timesteps at regular intervals from synchronised video (2688×1520 resolution, resized per model requirements). For each model, we:

1. Run inference on all timesteps (8 cameras per timestep)
2. Aggregate per-timestep poses via quaternion averaging (rotations) and median (translations)
3. Apply Sim(3) Procrustes alignment to ground truth (with reflection check)
4. Compute ATE, ARE, RPE, and co-located pair distances

Ground truth poses are obtained from COLMAP applied to a static scene capture (no cattle present). Two camera pairs (C2–C3 and C5–C6) share physical mounting poles, providing a structural validation metric (ground truth distance ≈ 0).

3.2 Results

3.3 Analysis

The results reveal a clear performance gap between architectures. Pi3X achieves sub-5° rotation error and 0.177 ATE—3.5× better in translation than the next model (MapAnything). MapAnything and VGGT form a second cluster at ∼10° rotation error: functional but substantially less

Table 1: Foundation geometry model benchmark on our wide-baseline, feature-adversarial dataset. Models are grouped by type (N-view joint vs. pairwise with post-hoc alignment) and ranked by ATE within each group. "Cond." indicates whether the model accepts pose/intrinsic priors as conditioning input—a prerequisite for our self-refinement method. All metrics use Sim(3)-aligned predictions; C2C3 measures co-located camera consistency. Pairwise models use only 20 timesteps due to computational cost.

Model	Venue	Type	Cond.	ATE↓	ARE↓	RPE_R↓	RPE_t↓	C2C3↓	Time (s)
N-view (joint)									
Pi3X	CVPR'25	N-view	✓	**0.177**	**4.37°**	**6.26°**	**0.39**	**0.088**	106
MapAnything	CVPR'25	N-view	✓	0.627	9.17°	14.19°	0.93	0.313	86
VGGT	CVPR'25	N-view	—	1.248	10.48°	12.82°	1.55	2.011	75
NoPo4D	arXiv'25	N-view	—	2.556	62.06°	77.01°	3.17	0.627	214
Fast3R	CVPR'25	N-view	—	2.743	96.64°	102.88°	4.36	3.183	460
Pairwise (post-hoc alignment)									
Pow3R	CVPR'25	Pairwise	✓	1.835	180°	28.53°	7.24	0.218	497
MonST3R	ICLR'25	Pairwise	—	2.607	47.24°	23.87°	3.56	0.111	1492

accurate. The remaining models (NoPo4D, Fast3R, MonST3R, Pow3R) all exceed 20° rotation error and effectively fail on this dataset.

Two structural patterns emerge from these results. First, N-view joint architectures (Pi3X, MapAnything, VGGT) consistently outperform pairwise models, confirming that joint multi-view reasoning is essential when inter-camera overlap is minimal—pairwise models cannot propagate geometric constraints across views that share no visual content. Second, the co-location metric reveals differences in structural understanding that are masked by aggregate error: Pi3X correctly infers that cameras C2 and C3 share a mounting pole (distance 0.088), while VGGT (2.011) and Fast3R (3.183) place them far apart.

Notably, Pi3X is one of only three models that support pose conditioning (Table **??**, "Cond." column)—a prerequisite for the self-refinement method developed in subsequent sections. Its combination of highest accuracy and conditioning support makes it the natural backbone. The remainder of this paper develops methods to push beyond Pi3X's $\sim$4.4° accuracy ceiling, which resists further improvement through standard geometric post-processing alone.

4 Method

Our method operates in four stages. Stage 1 applies standard geometric post-processing to establish a baseline estimate. Stages 2–4 constitute our contribution: self-conditioned iterative refinement with decoupled rotation-translation fusion.

4.1 Preliminaries: Geometric Post-Processing (Stage 1)

Given $N_t = 91$ timesteps of N-camera images, we run Pi3X to obtain per-timestep poses $\{T_k^c\}_{k=1}^{N_t}$ for each camera $c \in \{1, \ldots, N\}$, where each $T_k^c \in \text{SE}(3)$ is defined up to a per-timestep Sim(3) ambiguity (i.e., each timestep's output coordinate frame is arbitrary in position, orientation, and scale). We aggregate these into a single estimate via three standard steps.

Gauge fixing. Since each timestep uses an arbitrarily chosen global frame, we align all timesteps to a reference via Sim(3) Procrustes [?]:

$$\{s^*, R^*, \mathbf{t}^*\} =_{s,R,\mathbf{t}} \sum_{c=1}^{N} \|sR\,\mathbf{p}_k^c + \mathbf{t} - \mathbf{p}_{\mathrm{ref}}^c\|^2 \tag{1}$$

where $\mathbf{p}_k^c \in \mathbb{R}^3$ denotes the position of camera c at timestep k, and $s \in \mathbb{R}^+$, $R \in \mathrm{SO}(3)$, $\mathbf{t} \in \mathbb{R}^3$ parametrise the Sim(3) transform.

Outlier rejection. For each camera, we project its N_t estimated positions onto the dominant 2D plane (via PCA on camera centres) and reject outliers whose in-plane distance exceeds 3× the Median Absolute Deviation (MAD). Inlier rotations are aggregated via the Fréchet mean on SO(3) [?]—the rotation minimising the sum of squared geodesic distances to all inlier samples.

Pose graph optimization. We extract pairwise relative poses $T_{ij} = (T^i)^{-1}T^j$ from inlier timesteps, computing the Fréchet mean rotation and mean translation for each camera pair (i, j). These relative poses serve as edges in a pose graph, weighted by the inverse variance of the per-timestep estimates (edges with higher consistency receive more weight). We then optimise globally via L_1 Weiszfeld rotation averaging on SO(3) [?] followed by translation averaging with hard co-location constraints for cameras sharing a physical mounting pole.

Residual error analysis. This pipeline reduces ATE by 10% (0.177 → 0.161) but leaves ARE unchanged (4.37° → 4.39°). The persistence of rotation error under robust averaging over 91 independent timesteps indicates that the residual is systematic model bias rather than sample noise. Since the scene provides weak geometric signal (approximately planar static structure, dynamic repetitive content), further classical aggregation cannot improve upon this estimate. This motivates the use of Pi3X's conditioning interface to inject prior geometric information, as developed in the following sections.

4.2 Self-Conditioned Inference (Stage 2)

Pi3X supports multimodal conditioning: given approximate poses $\hat{T}^c$ and camera intrinsic matrices $\mathbf{\Pi}^c \in \mathbb{R}^{3\times3}$, it refines its predictions by attending to the provided geometric priors during encoding. We condition on the Stage 1 output:

$$\{T_k^{c,(1)}\} = f_\theta(\mathrm{imgs}_k \mid \hat{T}^c, \mathbf{\Pi}^c), \quad k = 1, \ldots, N_t \tag{2}$$

where f_θ denotes Pi3X and $\hat{T}^c$ is the Stage 1 estimate converted to the model's input camera ordering.

Applying the same geometric post-processing to $\{T_k^{c,(1)}\}$ yields a refined estimate with ARE reduced from 4.39° to 3.49° (20% improvement) but ATE increased from 0.161 to 0.166 (3% degradation). Iterating this process naïvely amplifies both effects: rotation improves further (to $\sim 3.3°) while translation degrades monotonically (to 0.195 ATE after 5 iterations). Section **??** provides a formal explar$ $the rotation component contracts toward a fixed point on SO(3), while translations diverge due to Sim(3) scale mismatc$

Algorithm 1 Iterated Hybrid MC Refinement

Require: Pipeline estimate $\hat{T}^c$, images, intrinsics $\mathbf{\Pi}^c$
Require: Noise: σ_R, σ_t; samples: S; iterations: M
$T_{\text{curr}} \leftarrow \hat{T}^c$
Compute unconditioned edges $\{\mathbf{t}_{ij}^{\text{uncond}}, w_{ij}^t\}$ {fixed throughout}
for $m = 1$ to M **do**
  for $s = 1$ to S **do**
    $T_s^c \leftarrow \text{Perturb}(T_{\text{curr}}^c, \sigma_R, \sigma_t)$ for all c
    $\{T_{k,s}^c\} \leftarrow f_\theta(\text{imgs} \mid T_s, \mathbf{\Pi}^c)$ for all k
    $\hat{T}_s^c \leftarrow \text{Pipeline}(\{T_{k,s}^c\})$
  end for
  Extract rotation edges $\{R_{ij}^{\text{cond}}, w_{ij}^R\}$ from pooled aligned poses
  Fuse: $\tilde{T}_{ij} = (R_{ij}^{\text{cond}}, \mathbf{t}_{ij}^{\text{uncond}})$
  $T_{\text{opt}} \leftarrow \text{PoseGraph}(\tilde{T}_{\text{init}}, \tilde{T}_{ij}, w_{ij})$
  $T_{\text{curr}} \leftarrow (R_{\text{opt}}, \mathbf{t}_{\text{uncond}})$ {anchor translations}
end for
return T_{opt}

4.3 Decoupled Hybrid Fusion (Stage 3)

Since conditioned inference improves rotation but degrades translation, we decouple them in the pose graph by sourcing edges from different inference runs. Let R_{ij} and $\mathbf{t}_{ij}$ denote the relative rotation and translation from camera i to camera j, computed as the mean over inlier timesteps. We construct fused edges:

$$\tilde{T}_{ij}^{\text{fused}} = \begin{pmatrix} R_{ij}^{\text{cond}} & \mathbf{t}_{ij}^{\text{uncond}} \\ \mathbf{0}^\top & 1 \end{pmatrix} \tag{3}$$

where R_{ij}^{cond} is the mean relative rotation extracted from conditioned inference (Stage 2) and $\mathbf{t}_{ij}^{\text{uncond}}$ is the mean relative translation from the original unconditioned run (Stage 1). Edge weights w_{ij} (inverse variance of the per-timestep relative pose estimates) are taken as $w_{ij} = \max(w_{ij}^{\text{cond}}, w_{ij}^{\text{uncond}})$.

The initial pose for graph optimization similarly uses conditioned rotations with unconditioned translations:

$$T_{\text{init}}^c = \begin{pmatrix} R_{\text{cond}}^c & \mathbf{t}_{\text{uncond}}^c \\ \mathbf{0}^\top & 1 \end{pmatrix} \tag{4}$$

This yields ARE = 3.45° (rotation improved) with ATE = 0.159 (translation preserved)—capturing the benefits of both sources.

4.4 Monte Carlo Iterative Refinement (Stage 4)

To further improve rotation estimates, we iterate the hybrid fusion with stochastic perturbation of the conditioning signal. The perturbation serves two purposes: (i) it explores the local neighbourhood of the current estimate, sampling multiple poses from the model's "denoising basin"; (ii) it provides diversity across ensemble members, reducing the variance of the averaged rotation edges.

The Perturb operation applies independent Gaussian noise to each camera's pose: rotation is perturbed via the exponential map $R' = R \cdot \exp([\boldsymbol{\epsilon}_R]_\times)$ where $\boldsymbol{\epsilon}_R \sim \mathcal{N}(\mathbf{0}, \sigma_R^2 I_3)$, and translation is perturbed additively $\mathbf{t}' = \mathbf{t} + \boldsymbol{\epsilon}_t$ where $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \sigma_t^2 I_3)$.

Design choices.

- **Noise magnitude**: $\sigma_R = 2°$ (axis-angle standard deviation), $\sigma_t = 0.05$ (isotropic, in scene units). These are chosen to stay within the model's contraction basin ($< 5°$; see Section **??**).
- **Translation anchoring**: Translations are *never* updated from conditioned outputs—only from the original unconditioned baseline. This is the key invariant that prevents Sim(3) drift.
- **Early stopping**: Terminate when ARE improvement between iterations falls below 0.1°, or after $M = 3$ iterations (empirically sufficient; see Section **??**).

Probabilistic interpretation. Each perturbed conditioned inference produces an approximate sample from the posterior $p(R \mid \text{imgs})$. Pooling $S \times N_t$ aligned poses across samples before extracting rotation edges estimates the posterior mean over relative rotations, reducing variance by $O(1/\sqrt{S})$. We formalise this connection to denoising score matching in Section **??**.

5 Theoretical Analysis

We provide a formal interpretation of the self-conditioning dynamics observed in Section **??**, explaining why rotation converges while translation diverges, and connecting the Monte Carlo perturbation strategy to established frameworks.

5.1 Contraction Mapping on SO(3)

Let $f : \mathrm{SO}(3)^N \to \mathrm{SO}(3)^N$ denote the composite mapping from conditioning rotations to Pi3X's output rotations after pipeline processing (gauge fix, outlier rejection, Fréchet mean). We define the geodesic distance on SO(3) as:

$$d_{\mathrm{SO}(3)}(R_1, R_2) = \| \log(R_1^\top R_2) \|_F \tag{5}$$

where $\log : \mathrm{SO}(3) \to \mathfrak{so}(3)$ is the matrix logarithm. This equals the angle of the relative rotation $R_1^\top R_2$.

If f is a contraction mapping, i.e., there exists $\lambda < 1$ such that for all R in a neighbourhood of the true rotations R^*:

$$d_{\mathrm{SO}(3)}(f(R), R^*) \leq \lambda \cdot d_{\mathrm{SO}(3)}(R, R^*) \tag{6}$$

then by the Banach fixed-point theorem, the iteration $R^{(n+1)} = f(R^{(n)})$ converges to a unique fixed point R^∞ satisfying $f(R^\infty) = R^\infty$.

Empirical measurement. We measure the contraction rate by feeding GT poses corrupted with known noise levels and comparing input error to output error (Section **??**, Table **??**):

- Input error 4.4° → output error 3.5° ($\lambda \approx 0.80$)
- Input error 5.0° → output error 3.3° ($\lambda \approx 0.67$)
- Input error 2.0° → output error 1.6° ($\lambda \approx 0.80$)

The contraction rate is approximately $\lambda \approx 0.8$ in the 2–5° regime. This predicts convergence to a fixed point at:

$$\text{ARE}^* = \frac{\lambda}{1-\lambda} \cdot \epsilon_{\text{irr}} \approx 4 \cdot \epsilon_{\text{irr}} \tag{7}$$

where ϵ_{irr} is the model's irreducible error (the residual even with perfect conditioning). With $\text{ARE}^* \approx 3.1°$ observed experimentally, this gives $\epsilon_{\text{irr}} \approx 0.8°$, consistent with the 1.2° achieved under perfect GT conditioning (which includes additional pipeline processing noise).

Basin of attraction. The contraction property holds only within a finite neighbourhood of R^*. Our noise robustness experiments (Section **??**) show that conditioning with $> 10°$ error causes catastrophic failure (output error jumps to 180°). This defines a basin of attraction of approximately $\pm 10°$ around the true solution—our initial estimate (4.4° error) lies safely within it.

5.2 Failure of Contraction for Translation

Pi3X outputs poses in Sim(3), not SE(3). The translation component $\mathbf{t}^c$ is meaningful only relative to the per-run scale s. When fed back as SE(3) conditioning, a scale mismatch accumulates:

$$\mathbf{t}^{(n+1)} = s^{(n)} \cdot g(\mathbf{t}^{(n)}, R^{(n)}) + \boldsymbol{\beta} \tag{8}$$

where $s^{(n)} \neq 1$ in general and $\boldsymbol{\beta}$ represents systematic bias from the conditioning mechanism. Since the scale factor $s^{(n)}$ varies per iteration and is not corrected, this does not satisfy contraction:

$$\|\mathbf{t}^{(n+1)} - \mathbf{t}^*\| \not\leq \lambda \|\mathbf{t}^{(n)} - \mathbf{t}^*\| \tag{9}$$

This analysis motivates our hybrid fusion strategy: rotation contracts and should be iterated; translation does not and should be anchored to the unconditioned estimate.

5.3 Monte Carlo Perturbation as Posterior Sampling

The stochastic perturbation in Stage 4 admits an interpretation through the lens of denoising score matching [**?**]. Under the hypothesis that Pi3X's conditioning mechanism was trained with noisy pose inputs (a common data augmentation for robustness), conditioned inference approximates the posterior mean:

$$f_\theta(R + \boldsymbol{\epsilon}) \approx \mathbb{E}[R^* \mid R + \boldsymbol{\epsilon}, \text{imgs}] \tag{10}$$

By running S perturbations $\{R \cdot \exp([\boldsymbol{\epsilon}_s]_\times)\}_{s=1}^S$ and computing the Fréchet mean of the outputs:

$$\bar{R} = \text{FréchetMean}\left(\{f_\theta(R \cdot \exp([\boldsymbol{\epsilon}_s]_\times))\}_{s=1}^S\right) \tag{11}$$

we approximate the marginal posterior mean $\mathbb{E}[R^* \mid \text{imgs}]$ via Monte Carlo integration over the perturbation distribution. The variance of this estimator decreases as $O(1/\sqrt{S})$.

Remark. This interpretation relies on the assumption that Pi3X was trained with noisy conditioning. While we cannot verify this from the published model, the empirical contraction behaviour (smooth error reduction rather than mode collapse) is consistent with a model that has learned to denoise its conditioning inputs. The interpretation should be understood as a motivating analogy rather than a proven equivalence.

5.4 Connection to Iterative Denoising

The overall procedure—starting from a noisy estimate, applying a learned map that moves closer to the truth, and iterating—shares the structure of denoising diffusion [?]:

1. Start with a noisy estimate $R^{(0)}$ (unconditioned Pi3X output, $\sim$4.4° error)
2. Apply a learned denoiser f_θ conditioned on images
3. The denoiser contracts the estimate toward the clean manifold
4. Iterate 2–3 times until convergence

The key differences from standard diffusion are: (i) we start close to the solution rather than from pure noise, requiring only 2–3 steps; (ii) we do not train a noise-level-conditional model; and (iii) convergence is restricted to the rotation subgroup, necessitating the hybrid decoupling. This suggests a broader principle: *pretrained geometry models with conditioning interfaces can serve as implicit denoisers at test time, even without explicit denoising training.*

6 Results

6.1 Experimental Setup

Dataset. We evaluate on 8 static cameras (2688×1520 resolution) deployed in a cattle feedlot. We sample $N_t = 91$ timesteps at regular intervals from synchronised video. Ground truth poses are obtained from COLMAP applied to a static capture of the same scene (without cattle).

Evaluation metrics. All methods are evaluated after Sim(3) Procrustes alignment to ground truth (with reflection check):

- **ATE**: Absolute Translation Error (RMSE of camera position errors)
- **ARE**: Absolute Rotation Error (mean geodesic distance to GT rotations, in degrees)
- **RPE rot / trans**: Relative Pose Error averaged over all $\binom{8}{2} = 28$ camera pairs
- **C2C3**: Distance between co-located cameras C2 and C3 (GT ≈ 0; structural consistency test)

Note that ATE and RPE trans are in arbitrary units (scene scale is determined by the Sim(3) alignment, not metric ground truth).

6.2 Comprehensive Comparison

Table **??** presents the progressive improvement from raw Pi3X output through our full method.

6.3 Analysis

The progressive results in Table **??** reveal several findings.

Rotation error is not reducible by classical aggregation. The pipeline (row 2) reduces ATE by 10% through pose graph optimization with co-location constraints, but provides no rotation improvement (4.37° → 4.39°). This confirms that Pi3X's rotation error is systematic model bias rather than statistical noise: robust averaging over 91 timesteps has already extracted all available information from the raw outputs.

Table 2: Progressive improvement from raw Pi3X through our full method. Cost is relative to a single unconditioned inference pass (∼100s on an A100). Row 3 shows naïve self-conditioning (rotation improves but translation degrades). Row 4 introduces hybrid fusion (translation preserved). Row 5 adds MC perturbation and iteration.

Method	ATE↓	ARE↓	RPE_R↓	RPE_t↓	C2C3↓	Cost
1. Raw Pi3X (quat average)	0.177	4.37°	6.26°	0.395	0.088	1×
2. + Pipeline (Stage 1)	0.161	4.39°	6.26°	0.375	0.000	1×
3. + Self-Conditioning (Stage 2)	0.166	3.49°	4.93°	0.384	0.000	2×
4. + Hybrid Fusion (Stage 3)	0.159	3.45°	4.93°	0.322	0.000	2×
5. + MC Iteration (Stage 4)	**0.160**	**3.13°**	**4.57°**	**0.311**	**0.000**	13×
Oracle (GT pose conditioning)	*0.097*	*1.22°*	—	—	—	1×

Self-conditioning exhibits rotation–translation asymmetry. Conditioning on the pipeline estimate (row 3) reduces ARE by 20% but increases ATE by 3%. This asymmetry—the central observation motivating our method—is consistent across multiple runs and noise configurations (Section **??**).

Hybrid fusion resolves the asymmetry. Decoupling rotation and translation channels (row 4) achieves 3.45° ARE (capturing the conditioned rotation improvement) while maintaining ATE at 0.159 (preserving the unconditioned translation quality). The computational overhead is minimal: one additional conditioned inference pass (2× total cost).

Monte Carlo iteration yields further rotation gains. The full method (row 5) reduces ARE by an additional 9% over single-step hybrid fusion (3.45° → 3.13°) through iterative perturbation and re-estimation. The 13× compute cost reflects $S = 4$ MC samples over $M = 2$ iterations across 91 timesteps—a cost–accuracy tradeoff appropriate for offline calibration scenarios.

Aggregate improvement. Comparing the raw model output (row 1) to the full method (row 5): ARE improves by 28% (4.37° → 3.13°), RPE rotation by 27%, RPE translation by 21%, and ATE by 10%. The co-location constraint (C2C3) is satisfied exactly under pose graph optimization (rows 2–5).

Oracle upper bound. The final row reports Pi3X conditioned on ground truth poses, representing the best achievable accuracy given perfect conditioning. This oracle achieves ARE = 1.22° and ATE = 0.097—reductions of 72% and 40% over the unconditioned baseline. The gap between our self-refined result (3.13°) and the oracle (1.22°) quantifies the remaining headroom attributable to conditioning quality. The monotonic improvement from unconditioned (4.39°) through self-conditioned (3.13°) to oracle (1.22°) confirms that the conditioning pathway provides genuine geometric information and that our method successfully extracts a substantial fraction of the available signal without external data.

7 Ablation Studies

We isolate the contribution of each design decision through four ablation experiments.

7.1 Effect of Hybrid Decoupling

Table ?? compares naïve iterative refinement (feeding the full conditioned output back without decoupling) against our hybrid approach.

Table 3: Naïve vs. hybrid iterative refinement. Naïve iteration improves rotation at the cost of translation; hybrid fusion achieves superior rotation improvement while preserving translation accuracy.

Method	ATE↓	ARE↓	ΔATE	ΔARE
Unconditioned baseline	0.161	4.39°	—	—
Naïve self-refine (iter 5)	0.195	3.60°	+21%	−18%
Hybrid MC (iter 2)	0.160	3.13°	−1%	−29%

Naïve iteration degrades ATE by 21% while improving ARE by only 18%. In contrast, hybrid fusion achieves 29% ARE improvement with negligible translation change. This confirms the theoretical prediction of Section ??: translation does not contract under self-conditioning and must be anchored to the unconditioned estimate.

7.2 Sensitivity to Conditioning Quality

To characterise the basin of attraction, we condition Pi3X with ground truth poses corrupted by varying levels of Gaussian noise (Table ??).

Table 4: Effect of conditioning pose quality on output accuracy. Translation noise is isotropic Gaussian in scene coordinates. Beyond ∼10° rotation noise, conditioning becomes counterproductive and the model diverges.

Rot. noise (°)	Trans. noise	ATE↓	ARE↓
0 (perfect GT)	0	0.097	1.22°
1	0.05	0.087	1.30°
2	0.10	0.080	1.60°
5	0.20	0.077	3.34°
10	0.50	0.368	180°
20	1.00	0.951	180°

The model tolerates conditioning noise up to approximately 5° of rotation error, producing output that improves upon the conditioning input. At 10°, the conditioning signal becomes misleading and the model collapses to degenerate solutions (180° ARE indicates random orientations). The transition is sharp rather than gradual, indicating a well-defined basin boundary. Our self-conditioned prior (∼4.4° error) lies near the edge of this basin, which explains both why self-refinement succeeds and why it saturates after 2–3 iterations as the estimate approaches the basin's contraction limit.

7.3 Convergence Behaviour

Table ?? reports per-iteration metrics for the full hybrid MC method.

The method converges monotonically through iterations 1–2, achieving a 29% ARE reduction. Iteration 3 shows slight regression, and iteration 4 produces catastrophic failure (orientation flip

Table 5: Convergence of iterated hybrid MC refinement ($S = 4$ samples per iteration). Iteration 2 achieves the minimum ARE; iteration 4 produces an orientation flip in the rotation estimate.

Iteration	ATE	ARE (°)	RPE_R (°)	RPE_t
0 (unconditioned)	0.161	4.39	6.26	0.375
1	0.162	3.43	4.88	0.321
2	0.160	3.13	4.58	0.309
3	0.163	3.38	4.70	0.307
4	0.160	180°	4.78	8.55

in at least one camera). This behaviour is consistent with the basin-of-attraction analysis: as the conditioned estimate approaches the $\sim 3°$ fixed point, the margin between the estimate's error and the basin boundary narrows, making the iteration sensitive to perturbation noise. We adopt a conservative stopping criterion of $M = 2$ iterations or ARE improvement below 0.1° between successive iterations.

7.4 Inefficacy of Sim(3) Re-alignment

One might hypothesise that translation degradation under naïve iteration is caused by scale drift—each iteration's output uses a different arbitrary scale, and this accumulates. To test this, we Sim(3)-align each iteration's output back to the conditioning frame before re-injection. Table **??** shows that this intervention has no measurable effect.

Table 6: Sim(3)-constrained vs. unconstrained naïve iteration. Constraining scale does not prevent translation degradation, confirming that the failure mechanism is systematic conditioning bias rather than scale drift.

Iter	Sim(3) ATE	Sim(3) ARE	Unconstrained ATE	Unconstrained ARE
0	0.152	4.69°	0.152	4.69°
1	0.160	3.50°	0.160	3.50°
2	0.180	3.31°	0.179	3.29°
5	0.197	3.61°	0.195	3.59°

The near-identical degradation trajectories confirm that translation failure is not a scale problem but a structural one: Pi3X's conditioning mechanism introduces systematic translational bias that is independent of the output's global scale. This motivates our decision to anchor translations entirely to the unconditioned source rather than attempting to correct the conditioned output.

8 Discussion

Practical deployment. The hybrid fusion strategy (Stage 3) provides the most favourable accuracy–cost tradeoff: a single conditioned inference pass (2× compute) yields ∼21% rotation improvement while preserving translation. The full MC iterative method (Stage 4, 13× compute) offers diminishing marginal gains and is appropriate primarily for offline applications where sub-3° rotation accuracy justifies the additional cost. In all cases, hybrid decoupling is essential—naïve iteration without it leads to progressive translation degradation regardless of the number of steps.

Generality of the framework. While our experiments use Pi3X, the decoupled hybrid refinement framework applies in principle to any learned geometry model that (i) accepts pose conditioning as input, and (ii) outputs poses defined up to a gauge ambiguity. These properties are shared by several recent architectures (Table **??**). The contraction analysis (Section **??**) provides a diagnostic: measuring input–output error reduction under corrupted conditioning reveals whether a given model exhibits the contraction property and, if so, at what rate. Models with $\lambda < 1$ are amenable to self-refinement; those with $\lambda \geq 1$ are not.

Limitations. Several constraints bound the applicability of our method. The accuracy floor ($\sim 3°$ ARE) is determined by the contraction rate and the model's irreducible error—it cannot be lowered without stronger external priors or architectural modifications to the conditioning mechanism. Translation accuracy is bounded by the quality of unconditioned inference, since hybrid fusion anchors translations to this source. The convergence is empirical: while iterations 1–3 consistently improve, iteration 4 and beyond can produce orientation flips (Table **??**), necessitating a stopping criterion. Finally, our evaluation is conducted on a single multi-camera installation; generalisation to other wide-baseline deployments with different camera counts and geometries remains to be validated.

Future work. Several directions merit investigation. First, Pi3X also accepts depth conditioning—monocular depth priors (e.g., Depth Anything v2) could provide complementary geometric signal without requiring ground truth poses, potentially breaking the rotation floor. Second, partial conditioning (conditioning only high-confidence cameras while leaving others unconditioned) may exploit the model's cross-view attention to propagate reliable pose information to uncertain views. Third, the fixed perturbation magnitude ($\sigma_R = 2°$) could be replaced by an adaptive noise schedule that decreases across iterations, following the annealing principle from diffusion-based methods. Fourth, sparse geometric anchors (a pre-calibrated camera pair or LiDAR registration points) could provide a stronger initialisation that places the estimate deeper within the contraction basin, accelerating convergence and lowering the fixed point.

9 Conclusion

We have shown that self-conditioning in learned geometry models exhibits a fundamental asymmetry: rotation contracts toward a fixed point on SO(3) while translation diverges due to Sim(3) scale ambiguity. Exploiting this asymmetry via decoupled hybrid fusion—sourcing rotation from conditioned inference and translation from unconditioned inference—yields consistent improvements without the degradation of naïve iteration. Combined with Monte Carlo perturbation, our method achieves 28% rotation improvement over the raw Pi3X baseline on a challenging wide-baseline, feature-adversarial calibration task, with no external calibration data required.

The contraction rate ($\lambda \approx 0.8$) and resulting accuracy floor ($\sim 3°$) are properties of the model and scene, not the method. Breaking this floor requires either stronger external priors or architectural changes to the conditioning mechanism. We believe the decoupled self-refinement framework generalises to any learned geometry model with a pose conditioning interface, and the contraction analysis provides a principled tool for predicting when iterative refinement will help.